

JEONGSEOB AHN

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EMPLOYMENT

Ajou University, Suwon, South Korea

Associate Professor in Department of Software and Computer Engineering Sep. 2021–Present

Assistant Professor in Department of Software and Computer Engineering Sep. 2017– Aug.2021

Research group: [Computer Systems Laboratory](#)

Oracle Labs, Belmont, CA February 2016 –July 2017

Researcher & Engineer

I participated in a research project building a HW accelerated big data analytics (called RAPID)

The University of Michigan, Ann Arbor, MI February 2015–January 2016

Research Fellow in Computer Science and Engineering

EDUCATION

KAIST, Daejeon, South Korea February 2011–February 2015

Ph.D. in Computer Science

Advisor: Professor Jaehyuk Huh

Dissertation: Architectural Supports for High Performance Address Translation and Processor Scheduling in Virtualized Systems

KAIST, Daejeon, South Korea February 2009–February 2011

M.S. in Computer Science

Advisor: Professor Jaehyuk Huh

Thesis: Scalable Subspace Snooping: Exploiting Temporal Sharing Stability

Dongguk University, Seoul, South Korea March 2004–February 2009

B.S. in Computer Science and Engineering

RESEARCH INTERESTS

Interested in all computer systems and architecture related problems in general. In particular, my research focus is the interaction of system software and architecture for building high performance computer systems. The following keywords indicates my interests: **Computer Architecture, Operating Systems, ML Systems, Virtualization, Cloud Computing, Datacenter Computing, and Performance Profiling.**

PUBLICATIONS

Conferences					Journals		
ISCA	MICRO	EuroSys	ATC	PACT	TC	TPDS	TACO
1	2	3	4	1	4	1	1

Table 1: Publications in top/major CS conferences and journals

The asterisk (*) denotes the role of the corresponding author after joining Ajou University (Sep. 2017).

Conference Proceedings

- [1] **EnvPipe: Performance-preserving DNN Training Framework for Saving Energy.**
Sangjin Choi, Inho Koo, Jeongseob Ahn, Myeongjae Jeon, and Youngjin Kwon.
In *Proceedings of the USENIX Annual Technical Conference (ATC)*, July 2023.
- [2] **Fast and Efficient Model Serving Using Multi-GPUs with Direct-Host-Access.**
Jinwoo Jeong, Seungsu Baek, and Jeongseob Ahn*.
In *Proceedings of the ACM European Conference on Computer Systems (EuroSys)*, May 2023.
* Received Gilles Muller Best Artifact Award.
- [3] **Memory Harvesting in Multi-GPU Systems with Hierarchical Unified Virtual Memory.**
Sangjin Choi, Taeksoo Kim, Jinwoo Jeong, Rachata Ausavarungnirun, Myeongjae Jeon, Youngjin Kwon, and Jeongseob Ahn*.
In *Proceedings of the USENIX Annual Technical Conference (ATC)*, July 2022.
- [4] **Rethinking Remote Memory Placement on Large-Memory Systems with Path Diversity.**
Wonkyo Choe, Sang-Hoon Kim, and Jeongseob Ahn*.
In *Proceedings of the ACM SIGOPS Asia-Pacific Workshop on Systems (APSys)*, August 2021.
- [5] **Exploring the Design Space of Page Management for Multi-Tiered Memory Systems.**
Jonghyeon Kim, Wonkyo Choe, and Jeongseob Ahn*.
In *Proceedings of the USENIX Annual Technical Conference (ATC)*, July 2021.
- [6] **Zico: Efficient GPU Memory Sharing for Concurrent DNN Training.**
Gangmuk Lim, Jeongseob Ahn, Wencong Xiao, Youngjin Kwon, and Myeongjae Jeon.
In *Proceedings of the USENIX Annual Technical Conference (ATC)*, July 2021.
- [7] **GrandSLam: Guaranteeing SLAs for Jobs in Microservices Execution Frameworks.**
Ram Srivatsa Kannan, Lavanya Subramanian, Ashwin Raju, Jeongseob Ahn*, Jason Mars, and Lingjia Tang.
In *Proceedings of the ACM European Conference on Computer Systems (EuroSys)*, March 2019.
- [8] **A Load Balancing Technique for Memory Channels.**
Byoungchan Oh, Nam Sung Kim, Jeongseob Ahn, Bingchao Li, Ronald Dreslinski, and Trevor Mudge.
In *Proceedings of the International Symposium on Memory Systems (MEMSYS)*, October 2018.
- [9] **Accelerating Critical OS Services in Virtualized Systems with Flexible Micro-sliced Cores.**
Jeongseob Ahn*, Chang Hyun Park, Taekyoung Heo, and Jaehyuk Huh.
In *Proceedings of the ACM European Conference on Computer Systems (EuroSys)*, April 2018.
- [10] **How Much Computation Power do you need for Near-Data Processing in Cloud?**
Namhyung Kim, Jeongseob Ahn, Sungpack Hong, Hassan Chafi, and Kiyoun Choi.
In *Workshop on Architectures and Systems For Big Data (ASBD)*, June 2017.
- [11] **Enhancing DRAM Self-Refresh Modes for Idle Power Reduction.**
Byoungchan Oh, Nilmini Abeyratne, Jeongseob Ahn, Ronald Dreslinski, and Trevor Mudge.

- In *Proceedings of the IEEE International Symposium on Low Power Electronics and Design (ISLPED)*, June 2016.
- [12] **Micro-sliced Virtual Processors to Hide the Effect of Discontinuous CPU Availability for Consolidated Systems.**
Jeongseob Ahn, Chang Hyun Park, and Jaehyuk Huh.
 In *Proceedings of the ACM/IEEE International Symposium on Microarchitecture (MICRO)*, December 2014.
 - [13] **Revisiting Hardware-Assisted Page Walks for Virtualized Systems.**
Jeongseob Ahn, Seongwook Jin, and Jaehyuk Huh.
 In *Proceedings of the ACM/IEEE International Symposium on Computer Architecture (ISCA)*, June 2012.
 - [14] **Dynamic Virtual Machine Scheduling in Clouds for Architectural Shared Resources.**
Jeongseob Ahn, Changdae Kim, Jaeung Han, Young-ri Choi, and Jaehyuk Huh.
 In *Proceedings of the USENIX Workshop on Hot Topics in Cloud Computing (HotCloud)*, June 2012.
 - [15] **Architectural Support for Secure Virtualization under a Vulnerable Hypervisor.**
 Seongwook Jin, **Jeongseob Ahn**, Sanghoon Cha, and Jaehyuk Huh.
 In *Proceedings of the ACM/IEEE International Symposium on Microarchitecture (MICRO)*, December 2011.
 - [16] **Subspace Snooping: Filtering Snoops with Operating System Support.**
 Daehoon Kim, **Jeongseob Ahn**, Jaehong Kim, and Jaehyuk Huh.
 In *Proceedings of the ACM/IEEE/IFIP International Conference on Parallel Architectures and Compilation Techniques (PACT)*, September 2010.
 - [17] **The Effect of Multi-core on HPC Applications in Virtualized Systems.**
 Jaeung Han, **Jeongseob Ahn**, Changdae Kim, Youngjin Kwon, Young-ri Choi, and Jaehyuk Huh.
 In *Workshop on Virtualization and High-Performance Cloud Computing (VHPC)*, August 2010.

Journal Articles

- [18] **Rethinking DRAM's page mode with STT-MRAM.**
 Byoungchan Oh, Nilmini Abeyratne, Nam Sung Kim, **Jeongseob Ahn**, Ronald Dreslinski, and Trevor Mudge.
IEEE Transactions on Computers (TC), 72(5), May 2023.
- [19] **Server Load and Network-aware Adaptive Deep Learning Inference Offloading for Edge Platforms).**
 Jungmo Ahn, Youngki Lee, **Jeongseob Ahn***, and JeongGil Ko*.
Elsevier Internet of Things (IoT), 21, April 2023.
- [20] **Reconciling Time Slice Conflicts of Virtual Machines with Dual Time Slice for Clouds.**
 Taeklim Kim, Chang Hyun Park, Jaehyuk Huh, and **Jeongseob Ahn***.
IEEE Transactions on Parallel and Distributed Systems (TPDS), 31(10), October 2020.
- [21] **A Study of Memory Placement on Hardware-assisted Tiered Memory Systems.**
 Wonkyo Choe, Jonghyeon Kim, and **Jeongseob Ahn***.
IEEE Computer Architecture Letters (CAL), 19(2), July - December 2020.
- [22] **LightCert: On Designing a Lighter Certificate for Resource-limited Internet of Things Devices.**
 Hyuksang Kwon, **Jeongseob Ahn**, and JeongGil Ko.
Transactions on Emerging Telecommunications Technologies (ETT), 30(10), September 2019.
- [23] **Caliper: Interference Estimator for Multi-tenant Environments Sharing Architectural Resources.**

Ram Srivatsa Kannan, Michael Laurenzano, **Jeongseob Ahn***, Jason Mars, and Lingjia Tang.
ACM Transactions on Architecture and Code Optimization (TACO), 16(3), May 2019.

- [24] **Deduplicating TLB Entries for Shared Pages.**
Jeongseob Ahn*.
IEICE Electronics Express (ELEX), 15(14), July 2018.
- [25] **Fast Two-level Address Translation for Virtualized Systems.**
Jeongseob Ahn, Seongwook Jin, and Jaehyuk Huh.
IEEE Transactions on Computers (TC), 64(10):3461–3474, December 2015.
- [26] **H-SVM: Hardware-assisted Secure Virtual Machines under a Vulnerable Hypervisor.**
 Seongwook Jin, **Jeongseob Ahn**, Jinho Seol, Sanghoon Cha, Jaehyuk Huh, and Seungryoul Maeng.
IEEE Transactions on Computers (TC), 64(10):2833–2846, October 2015.
- [27] **Subspace Snooping: Exploiting Temporal Sharing Stability for Snoop Reduction.**
Jeongseob Ahn, Daehoon Kim, Jaehong Kim, and Jaehyuk Huh.
IEEE Transactions on Computers (TC), 61(11):1624–1637, November 2012.

Granted Patents

- [28] **Apparatus and method for accelerating critical service in virtualized system.**
Jeongseob Ahn*, Jaehyuk Huh, Chang Hyun Park, and Taekyoung Heo.
Patent No.102115738, May 2020.
- [29] **System and method of reducing traffic among multi-cores used to meet cache coherence.**
 Jaehyuk Huh, **Jeongseob Ahn**, Daehoon Kim, and Jaehong Kim.
Patent No.1012331090000, February 2013.

TEACHING EXPERIENCE

Instructor at Ajou University

Semester Year	Course Number	Course Title	Enrollment	Course Rating
Spring 2023	SCE212	Computer Organization and Architecture (Undergraduate)	108	4.75/5
Fall 2022	SCE212	Computer Organization and Architecture (Undergraduate)	104	4.60/5
Spring 2022	SCE213	Operating Systems (Undergraduate)†	96	4.54/5
Fall 2021	SCE394	IT Intensive Programming (Undergraduate)†	24	4.59/5
Spring 2021	CSE561	Advanced Computer Architecture (Undergraduate & Graduate)*†	18	4.48/5
Spring 2021	SCE212	Computer Organization and Architecture (Undergraduate)†	106	4.58/5
Fall 2020	CSE812	Special Topics in Systems and Applications (Graduate)*†	12	4.70/5
Fall 2020	SCE212	Computer Organization and Architecture (Undergraduate)†	112	4.53/5
Spring 2020	SCE212	Computer Organization and Architecture (Undergraduate)†	104	4.48/5
Fall 2019	SCE212	Computer Organization and Architecture (Undergraduate)	102	4.59/5
Fall 2019	CSE561	Advanced Computer Architecture (Graduate)*	20	4.50/5
Spring 2019	SCE212	Computer Organization and Architecture (Undergraduate)	116	4.37/5
Fall 2018	SCE212	Computer Organization and Architecture (Undergraduate)	110	4.37/5
Fall 2018	CSE515	Advanced Operating Systems (Graduate, w/ JeongGil Ko)*	17	4.30/5
Spring 2018	SCE213	Operating Systems (Undergraduate)	53	4.35/5
Spring 2018	ICT331	Computer Organization (Undergraduate)	28	3.70/5
Fall 2017	SCE212	Computer Organization and Architecture (Undergraduate)*	62	4.08/5
Fall 2017	CSE561	Advanced Computer Architecture (Graduate)*	14	4.58/5

Table 2: The asterisk (*) denotes the lectures offered by English and dagger (†) means that the courses were provided by online due to COVID-19.

Teaching Assistant at KAIST

CS230 System Programming	Fall 2014, Fall 2011
CS310 Embedded Computer Systems	Fall 2012
CS510 Computer Architecture	Spring 2012, Spring 2010
CS311 Computer Organization	Spring 2011, Fall 2009

HONORS AND AWARDS

Gilles Muller Best Artifact Award at ACM EuroSys	2023
Research Excellence Award for 2022 at Ajou University	2023
Teaching Excellence Award for 2019 at Ajou University	2020
Outstanding Dissertation Award for 2015 in Computer Science at KAIST	2015
National Full Scholarship for Graduate Study	February 2009 - February 2015
Outstanding Teaching Assistant Award at KAIST	Fall 2012
Academic Scholarship at Dongguk University	Fall 2017, Spring 2018

PROFESSIONAL ACTIVITIES

Program Committee

IEEE International Symposium on High-Performance Computer Architectur (HPCA)	2024
IEEE International Symposium on Performance Analysis of Systems and Software (ISPASS)	2021

Conference External Reviewer

ACM/IEEE International Symposium on Computer Architecture (ISCA)	2019
IEEE International Symposium on High-Performance Computer Architecture (HPCA)	2016, 2013
ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)	2016
IEEE/ACM International Symposium on Microarchitecture (MICRO)	2015

Journal Reviewer

IEEE Computer Architecture Letters (CAL)	2022
ACM Transactions on Computer Systems (TOCS)	2019, 2021, 2022
IEEE Transactions on Parallel and Distributed Systems (TPDS)	2012, 2021
IEEE Transactions on Computers (TC)	2020, 2021
ACM Transactions on Embedded Computing Systems (TECS)	2019
IEEE Micro	2019, 2020, 2022
ACM Transactions on Design Automation of Electronic Systems (TODAES)	2016
IEEE Transactions on Cloud Computing (TCC)	2015

RESEARCH GRANTS

Building High-Performance AI Inference Serving Systems

Agency/Company: National Research Foundation of Korea (NRF)
Total Amount: 366,150,000 KRW
Role: PI
Period: March 2023 - February 2026
Share: 100%

Research on CXL-based Memory-Centric Server Systems

Agency/Company: SK Hynix
Total Amount: 90,000,000 KRW
Role: PI
Period: January 2021 - December 2022
Share: 100%

Morphable Memory Hierarchy for Supporting Diverse Heterogeneous Memory in Clouds

Agency/Company: National Research Foundation of Korea (NRF)
Total Amount: 340,000,000 KRW
Role: PI
Period: March 2019 - February 2022
Share: 100%

Ajou DREAM Artificial Intelligence Division for Educating Innovative Leaders (BK21 4th)

Agency/Company: National Research Foundation of Korea (NRF)
Role: Co-PI
Period: September 2020 - August 2027

IT Research Center for VR/AR-based Intelligent Life Consultants (ITRC)

Agency/Company: Institute for Information communication Technology Planning (IITP)
Role: Co-PI
Period: July 2021 - December 2028

Research on Container Scheduling Techniques for Achieving Low Latency in Edge Servers (3rd)

Agency/Company: Electronics and Telecommunications Research Institute (ETRI)
Total Amount: 60,000,000 KRW
Role: PI
Period: January 2022 - December 2022
Share: 100%

Research on Container Scheduling Techniques for Achieving Low Latency in Edge Servers (2nd)

Agency/Company: Electronics and Telecommunications Research Institute (ETRI)
Total Amount: 40,000,000 KRW
Role: PI
Period: January 2021 - December 2021
Share: 100%

Research on Container Scheduling Techniques for Achieving Low Latency in Edge Servers (1st)

Agency/Company: Electronics and Telecommunications Research Institute (ETRI)
Total Amount: 30,000,000 KRW
Role: PI
Period: August 2020 - November 2020
Share: 100%

Research on Scalable and Efficient DNN Serving System for Heterogeneous Servers

Agency/Company: NAVER Clova
Total Amount: 60,000,000 KRW
Role: PI
Period: June 2020 - November 2020
Share: 100%

Exploring Convergence Techniques for Future Cloud Computing

Agency/Company: Ajou University
Total Amount: 30,000,000 KRW
Role: PI
Period: October 2017 - September 2019
Share: 100%

Resource Management Techniques for Cloud Computing

Agency/Company: National Research Foundation of Korea (NRF)
Total Amount: 60,000,000 KRW
Role: PI
Period: September 2017 - August 2019
Share: 100%

Research on Highly Efficient Data Analytics Platform for Large-Scale Machine Learning

Agency/Company: National Research Foundation of Korea (NRF)
Total Amount: 13,550,000 KRW
Role: PI
Period: November 2017 - February 2018
Share: 100%

INVITED TALKS

Serving DNNs on GPU Servers in the Face of Cold Starts

SIGARCH Korea, Seoul, South Korea July 2023
ETRI, Daejeon, South Korea June 2023

Memory Harvesting in Multi-GPU Systems with Hierarchical Unified Virtual Memory

Korea University, Seoul, South Korea April 2023
NVRAMOS 2022, Sokcho, South Korea October 2022

Operating System Support for Large Memory Server

POSTECH, Pohang, South Korea October 2022

Designing Server Systems for Datacenters

Yonsei University, Seoul, South Korea May 2022

Fast and Efficient Model Serving on GPUs with Zero-Copy

ETRI, Daejeon, South Korea March 2022
NAVER Clova, Seongnam, South Korea June 2021

Operating System Support for Multi-Tiered Memory: A Case Study of Intel's Optane DCPMM (Online)

NVRAMOS 2021, South Korea October 2021

Research on Large-Scale Memory Systems at Ajou Univ. (Online)

ETRI, South Korea	September 2021
Memory Management for Tiered Memory Systems (Online)	
ETRI, South Korea	October 2020
Software Techniques for Improving Throughput and QoS in Cloud Computing	
ETRI, Daejeon, South Korea	June 2020
Improving Throughput of DNN Serving Systems (Online)	
NAVER Clova, South Korea	April 2020
Cross-layer System Design for Cloud Computing	
Korea University, Seoul, South Korea	Nov 2019
Challenges and Opportunities in the Era of Heterogeneous Memory Systems	
ETRI, Daejeon, South Korea	December 2018
Optimizing Consolidated Systems with Flexible Micro-sliced Cores	
UNIST, Ulsan, South Korea	September 2018
ETRI, Daejeon, South Korea	May 2018
Seoul National University, Seoul, South Korea	May 2018
Flexible Micro-sliced Cores for Virtualized Systems	
KIISE Computer System Society, Weonju, South Korea	January 2018
Architectural Support for High Performance Cloud Computing	
Ajou University, Suwon, South Korea	May 2017
Architectural Support for High Performance Virtualized Systems	
ARM Research, Austin, TX	November 2015
IBM Austin Research Lab, Austin, TX	November 2015
VMware Research Group, Palo Alto, CA	October 2015
System Virtualization and Micro-sliced Virtual Processors	
The University of Michigan, Ann Arbor, MI	March 2015

STUDENT SUPERVISION

Graduate Students

Jonghyeon Kim (Ph.D. program)
Jinwoo Jeong (Ph.D. program)
Seungsu Baek (M.S. program)
WooHyung Choi (M.S. program)

Alumni

Jungmo Ahn (w/ Prof. JeongGil Ko) (Ph.D. 2022), Hyundai MOBIS
Wonkyo Choe (M.S. 2021), Ph.D. program in Computer Science, University of Virginia
Ji-Eun Han (B.S. 2021), M.S. program in Artificial Intelligence, Ajou University

Taeklim Kim (B.S. 2020), Ph.D. program in Computer Science, University of Texas at Austin
Jinwoo Hwang (B.S. 2020), Ph.D. program in Computer Science, KAIST
Jiwon Jeon (B.S. 2020), AI System & Software Engineer, Rebellions

Last updated: July 21, 2023